

Muhammad Ali Hassan

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EDUCATION

University of Illinois Chicago

Master of Science in Computer Science | Fulbright Scholar

Sep 2026 – Jun 2028

Lahore University of Management Sciences (LUMS)

Bachelor of Science in Computer Science | Dean's Honor List, Merit Scholarship Recipient

Sep 2021 – May 2025

- Coursework: Data Structures, Algorithms, Operating Systems, Distributed Systems, Databases, Blockchain, Deep Learning
- Head Teaching Assistant for Blockchain, CS 100, and Discrete Maths.

EXPERIENCE

Tibbling Technologies | Neuro AI startup

Remote – Seattle

AI Engineer

Jan 2025 – Aug 2026

- Built a 200K parameter 1D segmentation model in PyTorch that cuts CPU inference time by 93% while exceeding Meta's SAM in Dice score across 36,000+ neuro-preclinical images, in preparation for a manuscript to Nature Methods.
- Generalized the model into a universal neuro-segmenter trained with self-supervised datasets containing neurons and axons, with CPU-only few-shot fine-tuning (DoRA/FiLM adapters) that adapts to a new imaging domain from one annotated patch.
- Shipped a full-stack Next.js app for in-browser fine-tuning and real-time 3D segmentation, validated by labs at the University of Washington and Zurich, with Broad Institute running colocalization analysis on data helping autism drug discovery.
- Migrated the Dockerized stack from AWS to a single commodity server by moving storage to OME-Zarr and inference to ONNX Runtime, eliminating recurring cloud costs while scaling to 100+ concurrent users on 100 GB+ 3D medical volumes.

Cloud Computing Research Lab, LUMS

Lahore, Pakistan

Research Assistant

Jun 2024 – May 2025

- Designed an LLM pipeline converting natural language into BPMN workflows for survival analysis, giving clinicians governed analytics via a MERN app, with structured prompting yielding near-perfect executable workflows across 17 types.
- Built a blockchain-based (Solidity, Ethereum) access-control system that authorizes dataset requests against XACML policies and logs an immutable audit trail on-chain, keeping patient data off-chain, published in IEEE OJ-CS (2026).

IoT Lab, LUMS

Lahore, Pakistan

Research Apprentice

Sep 2024 – Jan 2025

- Shipped a 5-camera PTZ network with WWF covering 300 km² of Northern Pakistan to catch forest fires, automating sweeps via Hikvision API with Python/OpenCV + YOLO inference that localizes hotspots at 99% accuracy for early authority alerts.

Center for Water Informatics & Technology

Lahore, Pakistan

Research Intern

Jun 2024 – Aug 2024

- Architected an ESP32-CAM monitor for floods and wildlife in Northern Pakistan, where conventional stations cost too much to deploy, screening frames on-device with Edge Impulse before GSM upload to server-side YOLOv5 at 1/10th the cost.

PROJECTS

Raft-Based Distributed Fault-Tolerant Key-Value Store | Go, RPC, Raft

- Implemented the Raft consensus protocol from scratch, including leader election, log replication, and crash-recovery persistence, which was validated against a test suite simulating network partitions, dropped RPCs, and node restarts.

Full Stack Hiring Platform (Link) | React, Node.js, Express, MongoDB, Socket.io

- Built asynchronous video interviews, real-time recruiter-candidate chat, and application tracking from posting through offer.

On-Device LLM Profiling | Python, Ollama, ADB, Android, LLMs

- Built a memory-tracing pipeline using Ollama and ADB to profile 2-bit quantized Llama 3.2 1B across low-cost Android devices, identifying KV-cache growth and cold-start latency as the primary bottlenecks limiting on-device inference.

PUBLICATIONS & AWARDS

- S. Ali, M. A. Hassan, H. Ahmad, A. Jai, B. Shafiq, et al. "Governance-Aware AI Workflow Infrastructure for Secure and Policy-Compliant Healthcare Analytics." **IEEE Open Journal of the Computer Society**, vol. 7, pp. 1377–1388, 2026
- Fulbright Master's Scholarship (2026) | Dean's Honor List, LUMS | Winner, Techstars Startup Weekend (2025)

LEADERSHIP

E-Gaming at LUMS – Co-Founder & General Secretary

- Led a 50+ member team running high-profile esports events with 1,000+ attendees and PKR 200K in sponsorships.

Project Indigenous – CEO and Co-Founder

- Pursued ASR/TTS for Pakistan's speech-tech gap, collected 10+ hours of Pashto speech, and won startup incubation.

SKILLS

Languages & ML: Python, C++, C, Go, TypeScript, JavaScript, Solidity, SQL, PyTorch, NumPy, OpenCV, LangChain

Infrastructure & Web: AWS, Docker, ONNX Runtime, Git, MongoDB, REST, React, Next.js, MERN, Napari, Socket.io